

The Algebraic Derivative and Applications

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The purpose of this essay is to combinatorially define a derivative operator which reproduces classical results and extends the foundations of differential geometry to rings other than $\mathbb{R}$ and $\mathbb{C}$. Familiarity with counting, linear algebra, and rings is assumed. Knowledge of multilinear algebra and geometry would be helpful but technically not required.

In chapter 1, after defining the basic properties of the derivative operator we investigate applications to factoring in extension rings. Chapter 2 is actually an appendix of results from Geometry and Clifford algebras that we will want. In Chapter 3 we begin a preliminary exploration into two different vector products, and we discuss some associated cochain complexes, as well as other miscellaneous results. Chapter 4 discusses the concept of tangents and intersection multiplicity. In Chapter 5 we discuss anti derivatives and prove Stokes theorem. The final chapter investigates transcendental functions and associated transcendental constants.

All rings R in this paper are commutative integral domains with 1_R a multiplicative identity, and all homomorphisms τ of rings $A \rightarrow B$ send $\tau(1_A) = 1_B$ multiplicative identities to multiplicative identities.

The reader is invited to compare this exposition to those typically describing concepts like *the formal derivative*, *the Hasse derivative*, *finite differences*, or *the q -derivative*.

In the past, people typically considered "the total change according to h " of a function, then in order to isolate the linear change they would consider the limiting quantity

$$f' = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

There are drawbacks in this definition. It is not valid in every commutative ring, and it obscures the role of higher order rates of change. Moreover it builds crutch intuitions about infinitesimals where they are not strictly necessary, which interferes with a straightforward explanation as to the meaning of an expression like df . In this essay I will not deny that this definition is valid in the fields where it is definable, but I hope to adequately address each of the identified deficiencies. In my opinion the formal derivative is bad pedagogy in response to these issues.

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1 The Differential Operator

1.1 A small amount of module theory

Given a ring R and a finite set $X = \{x, y, z, \dots\}$ we can consider $R[x, y, z, \dots]$ the module created by R adjoined the elements of X , whose elements are (finite) formal linear combinations of (finite) formal commutative products of the elements in X , aka polynomials. Arithmetic and degree are defined in the usual way. This module is finitely generated and free so therefore it has a basis, in this case the usual standard basis of monomials suffices. The reason we care is we want to be able to express linear transformations between polynomials as matrices no matter the underlying ring of coefficients. We will freely allow ourselves to suppress the reliance on X for the arguments of polynomials.

We will also want to define the **R -binomial coefficients** $\binom{n}{k}_R$ using the usual recurrence relation. Namely we say that $\binom{n}{0}_R = \binom{n}{n}_R = 1_R$, as well as $\binom{n}{k}_R = \binom{n-1}{k-1}_R + \binom{n-1}{k}_R$.

Exercise 1 Show that there is always a unique homomorphism $\bar{\pi} : \mathbb{Z} \longrightarrow R$ which has the property that $\bar{\pi}\binom{n}{k} = \binom{n}{k}_R$. For brevity we will omit the R subscript.

For now we will briefly limit ourselves to the case that $X = \{x\}$.

1.2 The Translation and Truncation Operators

The most basic linear operator which everyone learns that acts on polynomials is the **evaluation map** $e_{x \rightarrow r} : R[x] \longrightarrow R$ defined for any $r \in R$ and $f = \sum_{i=0}^{\deg(f)} a_i x^i$ as $e_{x \rightarrow r} f = f(r) = \sum_{i=0}^{\deg(f)} a_i r^i$. If we represented polynomials as a column vector, then the matrix in the standard basis for this linear transformation is a diagonal with consecutive powers of r for entries, starting at 1_R . That is not too bad, but we will want to investigate more complicated matrices.

As a generalization of the evaluation map, we also can define the **composition by g operator** $e_{x \rightarrow g(x)} : R[x] \longrightarrow R[x]$ for $g \in R[x]$ by the formula

$$e_{x \rightarrow g(x)} f = \sum_{i=0}^{\deg(f)} a_i g^i.$$

Writing down the matrix for this transformation is not the simplest task, one application of our future results will be to compute this matrix with relatively less effort than a "brute force" combinatorial argument. Instead we will want to focus on a special case within our reach. Define the **translate by r operator** $E_r : R[x] \longrightarrow R[x]$ as $E_r f = e_{x \rightarrow x+r} f = f(x+r)$.

Though it is not the subject of this paper, we should highlight that the **scale by q linear transformation** $M_q = e_{x \rightarrow qx}$ plays a similar role to E_r in q-Calculus.

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Exercise 2 Represent $f = \sum_{i=0}^{\deg(f)} a_i x^i$ as a column vector $(a_0, a_1, a_2, \dots)^T$, and write down the matrix form of E_r . Show that the matrix form of E_r over $R = \mathbb{Z}$ looks like

$$e_r = \begin{bmatrix} 1 & r & r^2 & \dots \\ 0 & 1 & 2r & \dots \\ 0 & 0 & 1 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

what is the form over any R ? Prove that for $k \in \mathbb{Z}$ we have $E_r^k = E_{\pi(k)r}$. Use this to write the matrix form of E_r^{-1} .

The translation operator lets us understand the total change of a function, but in order to do calculus we would like to be able to isolate changes by degree. To this end, for K a nonnegative integer, we define the k^{th} **truncation operator** $T_x^k : R[x] \rightarrow R[x]$ by $T_x^k f = \sum_{i=0}^{\min(\deg(f), k)} a_i x^i$ which has the effect of lopping off all the terms of x with power greater than k from f . If we had a different basis consisting of powers of a linear term for $R[x]$, say $1, x-r, (x-r)^2, \dots$ such that $f = \sum_{i=0}^{\deg(f)} b_i (x-r)^i$, we could instead define the k^{th} **truncation operator with respect to $x-r$** as $T_{x-r}^k : R[x] \rightarrow R[x]$ such that $T_{x-r}^k f = \sum_{i=0}^{\min(\deg(f), k)} b_i (x-r)^i$.

Going even further, note that in $R[x]$ by definition we can always write an element of $R[x]$ as a linear combination of powers of x , but if there is some other element $r \in R[x]$ such that for every $f \in R[x]$ we can write it as $f = \sum_{i=0}^{\infty} a_i r^i$ a linear combination of powers of r with only finitely many nonzero $a_i(x)$, then we can extend T_x^k to T_r^k by instead $f = \sum_{i=0}^k a_i r^i$ lopping off powers of r .

Exercise 3 Prove that the truncation operator is a projection $T_x^k T_x^k = T_x^k$ and write down the matrix for T_x^k with respect to the standard basis. Show that $T_{x-r}^k = E_{-r} T_x^k E_r$.

1.3 Differentials

We define dx to be a new adjoined symbol representing a fixed change in x , and the k^{th} **differential** of a polynomial as the output of an operator $d_k : R[x] \rightarrow R[x]$ according to $d_1 x = dx$ for some arbitrary nonzero $dx \in R$, and $d_k f = [T_{dx}^k - T_{dx}^{k-1}] E_{dx} f$ for all other functions.

Some back of the napkin math should convince you that in $R = \mathbb{Z}$ we have that $d_k x^n = \binom{n}{k} x^{n-k} dx^k$. We refer to d_1 as the linear differential. Notice that $d_k x^n = \binom{n}{k} x^{n-k} dx^k$ implies as special cases that $d_0 x^n = x^n$, as well as $d_1 x = dx$. For this section we will focus on proving the properties of d_1 and we will wait to discuss the more general properties of d_m till the a future subsection.

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Exercise 4 Show that as a matrix the linear differential looks like

$$d_1 = [T_{dx}^1 - T_{dx}^0]E_{dx} = \begin{bmatrix} 0 & dx & 0 & 0 & \dots \\ 0 & 0 & 2dx & 0 & \dots \\ 0 & 0 & 0 & 3dx & \dots \\ 0 & 0 & 0 & 0 & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix} = \left[\binom{j}{i} dx^{j-i} \mathbb{1}(j-i-1) \right]$$

where $\mathbb{1}(n)$ is an indicator function that is 1 when $n = 0$ and 0 otherwise. Find the similar matrix expression for d_k .

From the definition, we can see that sums of consecutive differentials create telescoping sums. For example we have that $(d_2 + d_1)f = [T_{dxc}^2 - T_{dx}^1]E_{dx}f + [T_{dxc}^1 - T_{dx}^0]E_{dx}f = [T_{dxc}^2 - T_{dx}^0]E_{dx}f$. From this pattern we have that the total change of a function can be written as a sum of all the changes in specific degrees, written explicitly as

$$(E_{dx} - I)f = \sum_{k=1}^{\infty} d_k f \Leftrightarrow f(x + dx) = d_0 f + d_1 f + \dots,$$

where we have identified If and $d_0 f$. Notice that this is always a finite sum for f a polynomial, and furthermore that $d_k f$ is always a multiple of $(dx)^k$. In practice we will want to apply differentials to differentials. So far nothing in our definition precludes this, but we need to proceed with care if we want to avoid one of the first initial pitfalls. To make this point clearer, let us first prove the first theorem of this paper.

Theorem 1 Powers of the differential are a factorial multiple of higher differentials

$$d_1^k = k! d_k.$$

Proof We will proceed by induction, this is obviously true for $k = 0, 1$. Now suppose it is true for $k = m$. Then we have

$$d_1 d_m = \left[\binom{j}{i} r^{j-i} \mathbb{1}(j-i-1) \right] \left[\binom{j}{i} r^{j-i} \mathbb{1}(j-i-m) \right]_{i,j=0}^n = [a_{ij}]$$

for values $a_{ij} = \sum_{l=0}^n \binom{i}{l} \binom{l}{j} r^{j-i} \mathbb{1}(j-i-m) \mathbb{1}(j-i-1)$. For given i, j, m the value of l which causes both of our indicator functions to be nonzero occurs when

$$j - l - m = 0, \quad l - i - 1 = 0 \quad \implies \quad j - i = m + 1$$

So that the only terms that survive are those on the sub-diagonal where $j - i = m + 1$ and we can rewrite our sums as the single terms

$$d_1 d_m = \left[\binom{j}{l} \binom{l}{i} r^{j-i} \mathbb{1}(j-i-m-1) \right]_{i,j=0}^n = \left[\frac{j!}{l!(j-l)!} \frac{l!}{i!(l-i)!} r^{j-i} \mathbb{1}(j-i-m-1) \right]_{i,j=0}^n$$

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Since we can write $j - l = m$ and $l - i = 1$ then our expression simplifies to

$$d_1 d_m = \left[\frac{j!}{l!(j-l)!} \frac{l!}{i!(l-i)!} r^{j-i} \mathbb{1}(j-i-m-1) \right]_{i,j=0}^n = \left[\frac{j!}{m!i!} r^{j-i} \mathbb{1}(j-i-m-1) \right]_{i,j=0}^n$$

If we divide and multiply by $(m+1)$ then use the substitution $m+1 = j-i$ we get

$$\begin{aligned} d_1 d_m &= \left[\frac{(m+1)i!}{(j-i)!i!} r^{j-i} \mathbb{1}(j-i-m-1) \right]_{i,j=0}^n = (m+1) \left[\binom{j}{i} r^{j-i} \mathbb{1}(j-i-m-1) \right]_{i,j=0}^n \\ &\implies d_1 d_m = (m+1) d_{m+1} \end{aligned}$$

which completes the induction. □

So what is the problem here? Since we developed the result using a basis and matrices, it seems undeniable that the n^{th} differential of an m^{th} differential is an $(m+n)^{\text{th}}$ differential times a constant. However, if we think about the intuitive idea of a truncation in dx as a way to lop off higher powers of dx , then should we have that $d_1 d_m$ is a multiple of dx or $(dx)^{m+1}$? While this question is definitively answered by the matrix computations we have performed above, we would like any approach we use to proving theorem 1 to give the same answer.

To be more concrete, consider the case of $d_1 d_m(x^n)$ for $m \neq 0$. Evaluating the first differential gives us $d_1 \left(\binom{n}{m} x^{n-m} (dx)^m \right)$. If we want to lop off all terms of dx beyond linear then we would conclude $d_1 \left(\binom{n}{m} x^{n-m} (dx)^m \right) = 0$. Similarly, we can consider the case $d_m d_1(x^n) = d_m(n x^{n-1} dx) = n \binom{n-1}{m} x^{n-1-m} (dx)^m$.

This is disastrous because the structure of the matrix forms of differentials suggests they are generated by d_1 and therefore should commute. The way that we resolve this and bring everything into accordance is by understanding that as linear transformations, T_{dx}^k and d_m are acting on the basis monomials x^n , and any powers of dx already present in our expression are simply coefficients to be pulled out by linearity. The reader is invited to check that this interpretation produces results consistent with our matrix computations.

Before we move on, we also would like to take the time to introduce one more linear operator, the m^{th} **derivative** $D_m : R[x] \rightarrow R[x]$ by $D_m = d_m / (dx)^m$. Theorem 1 tells us that we can write $D_m = D_1^m / m!$, which will let us touch base with classical results. Our interpretation of an m^{th} differential is that it is the m^{th} of change in a function proportional to the m^{th} power of change in the input, whereas the derivative is that it is the m^{th} of change in a function independent to the m^{th} power of change in the input. In some sense the m^{th} derivative is the "instantaneous" degree m change of the function. As a consequence of the definition, we can also write $d_m f(x) = D_m f(x) (dx)^m$, and interpret the m^{th} derivative $D_m f$ as the coefficient of dx^m in $d_m f$.

One of the most important applications of classical derivatives is the Taylor expansion. In order to understand local information about a curve we will want to rewrite a curve from being expressed in terms of linear combinations of $x, y, x^2, xy, y^2, \dots$ to linear combinations of $x-a, y-b, (x-a)^2, (x-a)(y-b), (y-b)^2, \dots$. In the case of a single variable, called **Taylor**

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series, we can use polynomial long division to compute the new coefficients, but this process is not always the most efficient or illuminating.

There are more proofs of the existence and uniqueness of Taylor series, which also explain the role of the derivative for simplifying computations. It is a fun challenge to use only the operators we have defined so far for this task, without explicitly writing out the components of the matrices. We will postpone a more conceptually simple approach till section 3.

For now we will provide an exercise that explains the role of the derivative for computing coefficients indicating local information, but also explains quite a bit more. Though the purpose of this essay is to examine an algebraic perspective of the continuous derivative, we have many opportunities to make contact with other forms of calculus like finite differences.

Exercise 5 *Taylor series using Umbral Calculus* First we need a couple of definitions. A linear operator Q on the space of polynomials is said to be a **shift invariant delta** operator if

1. $QE_r = E_rQ$ the operator commutes with translation,
2. $Qr = 0$ the operator sends all constants to zero,
3. for $p(x)$ of degree n , we have that $Qp(x)$ is of degree $n - 1$.

A sequence of polynomials $p_n(x)$ is said to be a **(convolutional) basic sequence at r for Q** if the following properties hold

1. $p_0(x) = 1$,
2. $p_n(r) = 0$ for $n > 0$,
3. $Qp_n = p_{n-1}$.

Show that for any shift invariant delta operator we can always construct a basic sequence at r for Q , say $p_n(x)$, and it will always have the following properties

1. $p_n(x + y) = \sum_{k=0}^n p_k(x)p_{n-k}(y)$,
2. for any polynomial $q(x)$ we have $q(x) = \sum_{k=0}^{\infty} Q^k q(r)p_k(x)$.

Conclude the existence of a Taylor series centered at r for any polynomial

$$f(x) = \sum_{k=0}^{\infty} \frac{D_1^k f}{k!}(r)(x - r)^k = \sum_{k=0}^{\infty} D_k f(r)(x - r)^k.$$

1.4 Algebraic properties of linear differentials

The purpose of this section is to establish the absolutely most basic properties of differentials, to make sure we are maintaining contact with what we would expect from classical results. Later we will extend these results to more general expressions. For now we want to state and prove the usual product, chain, quotient, and inverse function rules for differentials. We can obtain the versions for derivatives by dividing by powers of dx .

Theorem 2 *Linear Product Rule* The linear differential of a product of functions is

$$d_1 \left[\prod_{k=0}^n f_k(x) \right] = \sum_{k=0}^n (d_1 f_k(x)) \prod_{\substack{k=0 \\ k \neq i}}^n f_k(x)$$

the sum of products with one function differentiated at a time.

Proof Using the definition of the linear differential we can say

$$d_1 \left[\prod_{k=0}^n f_k(x) \right] = [T_{dx}^1 - T_{dx}^0] E_{dx} \prod_{k=0}^n f_k(x) = [T_{dx}^1 - T_{dx}^0] \prod_{k=0}^n f_k(x + dx)$$

then by using the differential expansion of a translated function we have

$$d_1 \left[\prod_{k=0}^n f_k(x) \right] = [T_{dx}^1 - T_{dx}^0] \prod_{k=0}^n \left(\sum_{i=0}^{\infty} d_i f_k(x) \right)$$

While this sum may look formidable, we are only interested in the terms that are linear in dx . The only way to get a term out of this product that is linear in dx is by choosing one function to be differentiated, and by taking the rest of the terms to be the original functions. $\square$

In particular theorem 2 implies that $d_1(f(x)g(x)) = (d_1 f(x))g(x) + f(x)(d_1 g(x))$ the usual product rule.

In many contexts, the product rule is considered the fundamental property that sets a derivative apart from a regular linear transformation. What we have shown is that for a suitably appropriate definition of instantaneous linear change, the product rule is a necessary consequence. We will see it replicated in every generalization of this topic we have discussed. It is an interesting exercise to convince yourself that the product rule is also a sufficient property for a linear transformation to have in order to qualify as a linear measurement of a rate of change.

The argument above can also be modified to discuss higher differentials of products, but we will cover that in the next session, and we will take a slightly different approach, which will connect to the classical result. Given a ring $R[x]$, we frequently want to work in its field of fractions $R(x)$, and we want to extend our differential operator to this setting.

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Theorem 3 Suppose $h \in R(x)$ is defined by $h(x) = f(x)/g(x)$ for $f(x), g(x) \in R[x]$. Then we have that

$$d_1 h(x) = \left[\frac{(D_1 f(x))g(x) - f(x)(D_1 g(x))}{g(x)^2} \right] dx.$$

Proof We can rewrite our relation from $h(x) = f(x)/g(x)$ to $g(x)h(x) = f(x)$. By taking the linear differential of both sides we get

$$d_1 f(x) = d_1(g(x)h(x)) = d_1 f(x) \implies (d_1 g(x))h(x) + g(x)d_1 h(x).$$

We can factor out a dx from both sides to write

$$(D_1 f(x))dx = [(D_1 g(x))h(x) + g(x)(D_1 h(x))]dx$$

Substituting $h(x) = f(x)/g(x)$ and solving for $D_1 h(x)$ achieves the result. □

Next we want to prove the power rule for linear differentials, but we will take a highly non traditional route and point out a lemma first.

Lemma 1 Linear differentials commute with function composition

$$d_1(f(g(x))) = (d_1 f) \circ (g(x)).$$

Proof Starting with the right side we have

$$d_1(f(g(x))) = [T_{dx}^1 - T_{dx}^0]E_{dx}(f(g(x))) = [T_{dx}^1 - T_{dx}^0](f(g(x+dx))) = [T_{dx}^1 - T_{dx}^0](f(\sum_{k=0}^{\infty} d_k g(x)))$$

Since we are looking for the terms that are linear in dx we can throw away $d_2 f(g(x)) + d_3 f(g(x)) + \dots$ and we get

$$d_1(f(g(x))) = [T_{dx}^1 - T_{dx}^0](f(T_{dx}^1 \sum_{k=0}^{\infty} d_k g(x))) = [T_{dx}^1 - T_{dx}^0](f(g(x) + d_1 g(x)))$$

We can break down $f(g(x) + d_1 g(x))$ into its differentials according to g and $d_1 g$ (which is linear in dx)

$$d_1(f(g(x))) = [T_{dx}^1 - T_{dx}^0] \sum_{i=0}^n (d_i f) \circ (g(x)) = (d_1 f) \circ (g(x))$$

which achieves the result. □

This is not a proof that derivatives commute with function composition, which is not true.

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Theorem 4 *Linear Chain Rule* If we take the linear differential of a composition of functions $f(g(x))$ we get

$$d_1(f(g(x))) = ((D_1f) \circ g)(D_1g)dx$$

the derivative of the outside evaluated at the inside times the differential of the inside times dx .

Proof Since $d_1f(g)$ is linear in d_1g we can divide and multiply $d_1(f(g(x))) = (d_1f) \circ (g(x))$ by d_1g and do the same trick again with dx giving us

$$d_1(f(g(x))) = (d_1f) \circ (g(x)) = \frac{d_1f(g)}{d_1g}d_1g = (D_1f \circ g(x))d_1g$$

$$(D_1f \circ g(x))d_1g(x) = (D_1f \circ g(x))\frac{d_1g(x)}{dx}dx = (D_1f \circ g(x))(D_1g(x))dx$$

as required. $\square$

Dividing by dx gives us the classical result for derivatives, and shows that derivatives do not commute with function composition. If confusion about this result persists, we should note that when we compose $d_1f = D_1fd_1x$ with $g(x)$ we need to substitute $g(x)$ in for the argument of d_1x as well as the argument of D_1f .

For the final result of this section we will establish the inverse function rule. Suppose that for a function f we have $g(x)$ a compositional inverse so that $f(g(x)) = x$, then we have the following result in the field of fractions for $R[x]$.

Theorem 5 *Inverse Function Rule* If $g(x)$ is a compositional inverse of $f(x)$ then we have that

$$d_1g(x) = \left[\frac{1}{D_1f \circ g(x)} \right] dx.$$

Proof Applying the chain rule applied to the relation $f(g(x)) = x$ achieves the result. $\square$

At this point we have established all the basic theorems of differentials. There are a lot of directions we could go in, but we will want to extend our results to multiple variables before diving into specific applications.

1.5 Partial derivatives

We now move onto the situation of $R[X]$ where $X = \{x_1, x_2, \dots, x_m\}$ is a finite set of multiple variables.

Exercise 6 Show that $R[x_1, \dots, x_m]$ is isomorphic to $R[x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_m][x_i]$. Conclude that any multivariate polynomial can be viewed as a single variate polynomial with

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coefficients that are expressions of the other variables.

To deal with the situation where we have multiple variables, we need to extend our notation, and the definition of our linear operators. Instead of writing $f(x_1, x_2, \dots, x_m)$, we will instead write $f(\mathbf{x})$, where we are thinking of the argument $(x_1, x_2, \dots, x_m) = \mathbf{x}$ as a vector in $R[X]$. If $\mathbf{y}$ is another vector in $R[X]$, then we can extend the definition of the composition by $\mathbf{y}$ operator as $e_{\mathbf{x} \rightarrow \mathbf{y}} f(\mathbf{x}) = f(\mathbf{y})$. Again we will be concerned with the special case of $e_{\mathbf{x} \rightarrow \mathbf{x} + \mathbf{v}} = E_{\mathbf{v}}$ translation by the constant vector $\mathbf{v}$.

Now let us be more specific about the form of elements in $R[x]$. For $f \in R[X]$, we can write it as

$$f = \sum_{n=0}^{\deg(f)} \sum_{i_1+i_2+\dots+i_k=n} a_{i_1, i_2, \dots, i_k} x_1^{i_1} x_2^{i_2} \dots x_k^{i_k},$$

but this notation is unwieldy, we should use something better. Define a multi-index to be a list $\alpha = (i_1, i_2, \dots, i_m)$ of m nonnegative integers. We write $\#\alpha = i_1 + i_2 + \dots + i_k$, $a_\alpha = a_{i_1, i_2, \dots, i_k}$ and $\mathbf{x}^\alpha = x_1^{i_1} x_2^{i_2} \dots x_k^{i_k}$, so that we have

$$f = \sum_{n=0}^{\deg(f)} \sum_{\#\alpha=n} a_\alpha \mathbf{x}^\alpha$$

where the second sum is over all lists of nonnegative integers whose elements sum to n , and a_α is the coefficient of the corresponding monomial $\mathbf{x}^\alpha$ (which is zero if that monomial is not present in f). We will also want to examine multivariate polynomials written in some other basis, and to that end let $\mathbf{r} = (r_1, r_2, \dots, r_k)$. We will extend our multi index notation by allowing

$$f = \sum_{n=0}^{\deg(f)} \sum_{i_1+i_2+\dots+i_k=n} b_{i_1, i_2, \dots, i_k} (x_1 - r_1)^{i_1} (x_2 - r_2)^{i_2} \dots (x_k - r_k)^{i_k} = \sum_{n=0}^{\deg(f)} \sum_{\#\alpha=n} b_\alpha (\mathbf{x} - \mathbf{r})^\alpha$$

with the same interpretation of the inner sum as before. Then we define $T_{\mathbf{x}}^k : R[X] \rightarrow R[X]$ by the formula

$$T_{\mathbf{x}}^k f = \sum_{n=0}^{\min(k, \deg(f))} \sum_{\#\alpha=n} a_\alpha \mathbf{x}^\alpha$$

as well as

$$T_{\mathbf{x}-\mathbf{r}}^k f = \sum_{n=0}^{\min(k, \deg(f))} \sum_{\#\alpha=n} b_\alpha (\mathbf{x} - \mathbf{r})^\alpha.$$

Exercise 7 Similar to the one variable case, show that $T_{\mathbf{x}-\mathbf{r}}^k = E_{-\mathbf{r}} T_{\mathbf{x}}^k E_{\mathbf{r}}$. What does this change of basis formula say about the existence of multivariable Taylor series? Use this formula to show that the linear truncation of $ax^2 + 2bxy + cy^2 = ar^2 + 2brs + cs^2$ at $\mathbf{x} = (r, s)$ is given by $arx + b(sx + ry) + csy = ar^2 + 2brs + cs^2$.

The Algebraic Derivative and Applications

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We define the **partial differential operator** $\partial_i : R[X] \rightarrow R[X]$ by $[T_{dx_i}^1 - T_{dx_i}^0]E_{\mathbf{e}_i dx_i}$ where $\mathbf{e}_i$ is the i^{th} standard basis vector. In accordance with the single variable case, we also define the i^{th} partial derivative by $D_i f = (\partial_i f)/dx_i \iff \partial_i f = (D_i f)dx_i$. When our variables are not indexed x, y then the notation ∂_x, ∂_y will be used.

Some back of the napkin math should be able to convince you that $\partial_i x_j = [\mathbb{1}(i - j)]dx_i$, and otherwise this operator behaves like our regular differentials where we view variables other than x_i as elements of $S = R[X - \{x_i\}]$ and our function living in $S[x_i]$. What is less clear is whether $\partial_i \partial_j = \partial_j \partial_i$ our partial differential operators commute with each other for $i \neq j$. Fortunately for the functions we are concerned with, they do.

Theorem 6 *Clairout's Theorem* For $f \in R(X)$ multivariate rational functions with coefficients in R , we have that $\partial_i \partial_j f(\mathbf{x}) = \partial_j \partial_i f(\mathbf{x})$.

Proof We will prove this in two stages. First we will show that it holds for $f \in R[x]$ polynomials, then we will show it hold for functions defined as a ratio of two other functions whose partial differentials commute.

In order to show that it holds for polynomials, we only need to demonstrate it for a generic basis element of $R[x_i, x_j]$, and the more general cases follow from there. To that end we write

$$\partial_i \partial_j (x_i^m x_j^n) = \partial_i (n x_i^m x_j^{n-1}) = mn x_i^{m-1} x_j^{n-1} = \partial_j (m x_i^{m-1} x_j^n) = \partial_j \partial_i (x_i^m x_j^n)$$

as required.

Now suppose that $\partial_i \partial_j f(\mathbf{x}) = \partial_j \partial_i f(\mathbf{x})$, $\partial_i \partial_j g(\mathbf{x}) = \partial_j \partial_i g(\mathbf{x})$, and $h(\mathbf{x}) = f(\mathbf{x})/g(\mathbf{x})$. Then by taking the first partial differential of $\partial_i \partial_j (h(\mathbf{x}))$ we have

$$\partial_i \partial_j \left(\frac{f(\mathbf{x})}{g(\mathbf{x})} \right) = \partial_i \left(\frac{(\partial_j f(\mathbf{x}))g(\mathbf{x}) - f(\mathbf{x})(\partial_j g(\mathbf{x}))}{g(\mathbf{x})^2} \right).$$

Then taking the second gives us

$$\partial_i \partial_j \left(\frac{f(\mathbf{x})}{g(\mathbf{x})} \right) = \frac{((\partial_i \partial_j f(\mathbf{x}))g(\mathbf{x}) + (\partial_j f(\mathbf{x}))(\partial_i g(\mathbf{x}) - (f(\mathbf{x})(\partial_j \partial_i g(\mathbf{x})))g(\mathbf{x})^2 + (\partial_j f(\mathbf{x}))g(\mathbf{x}) - f(\mathbf{x})(\partial_j \partial_i g(\mathbf{x})(2g(\mathbf{x})))}{g(\mathbf{x})^4}.$$

We could clean this up but why bother when we really just want to invoke the fact that $\partial_i \partial_j f(\mathbf{x}) = \partial_j \partial_i f(\mathbf{x})$ and $\partial_i \partial_j g(\mathbf{x}) = \partial_j \partial_i g(\mathbf{x})$, so that

$$\partial_i \partial_j \left(\frac{f(\mathbf{x})}{g(\mathbf{x})} \right) = \frac{((\partial_j \partial_i f(\mathbf{x}))g(\mathbf{x}) + (\partial_j f(\mathbf{x}))(\partial_i g(\mathbf{x}) - (f(\mathbf{x})(\partial_j \partial_i g(\mathbf{x})))g(\mathbf{x})^2 + (\partial_j f(\mathbf{x}))g(\mathbf{x}) - f(\mathbf{x})(\partial_j \partial_i g(\mathbf{x})(2g(\mathbf{x})))}{g(\mathbf{x})^4}$$

which we can factor out a ∂_j out of to get

$$\partial_i \partial_j \left(\frac{f(\mathbf{x})}{g(\mathbf{x})} \right) = \partial_j \left(\frac{(\partial_i f(\mathbf{x}))g(\mathbf{x}) - f(\mathbf{x})(\partial_i g(\mathbf{x}))}{g(\mathbf{x})^2} \right)$$

which is $\partial_j \partial_i (f(x)/g(x))$ as required. □

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At this point a reader who remembers something about their Calculus 3 class may object that they recall pathological behavior identified by Peano from the function

$$f(x, y) = \begin{cases} \frac{xy(x^2-y^2)}{x^2+y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

which looks very reasonable if you graph it, but depending on the path you take, you could get different values for the second differentials. Specifically, he noted that

$$e_{x \rightarrow 0} \partial_x e_{y \rightarrow 0} \partial_y f(x, y) = dxdy$$

whereas if we do the partial differentials the other way around we get

$$e_{y \rightarrow 0} \partial_y e_{x \rightarrow 0} \partial_x f(x, y) = -dxdy$$

the piecewise definition of this function could not save us from the pathology at the origin. So whats wrong here? How does this not contradict the algebra in the proof above? Well here is the problem. Using the fact that differentials commute with function composition, and that $e_{x \rightarrow 0} e_{y \rightarrow 0} = e_{(x,y) \rightarrow (0,0)}$ we can observe that

$$e_{x \rightarrow 0} \partial_x e_{y \rightarrow 0} \partial_y f(x, y) = e_{y \rightarrow 0} \partial_y e_{x \rightarrow 0} \partial_x f(x, y) = e_{(x,y) \rightarrow (0,0)} \partial_x \partial_y f(x, y) = \frac{0}{0}$$

which is either undefined, or we have to allow every number to be equal to every other number. So the issue here is that we are trying to evaluate a partially defined map at one of it's singularities, which is never allowed in the first place.

Exercise 8 Let R be our a ring as well as $X = \{ x_1, x_2, \dots, x_m \}$ and $B = \{ b_1, b_2, \dots, b_n \}$ be finite sets. Denote the set of all $R[X]$ -linear combinations of B by $R[X]B$. Show that $R[X]B \subseteq R[X, B]$ and that $\mathbb{F}[X]B$ is a vector space when $\mathbb{F}$ is a field. Conclude that we can take the (partial) differential of any expression in $R[X]B$ with respect to an element of X by considering our expression as an element of $R[B][X]$ so that

$$\partial_i()$$

With Clairout's theorem out of the way, we have one more critical topic to discuss. We will want to extend the chain rule to the composition of maps of the form $f(\mathbf{x}) : R[t]B \rightarrow R[t]$ and $\mathbf{v}(t) : R[t] \rightarrow R[t]B$.

1.6 Properties of Higher Differentials

1.7 Factoring and Resultants

1.8 More on Ring Extensions

2 Appendix: Geometry Review

2.1 Affine and Projective spaces

2.2 Vector Products

3 Vector Derivatives

3.1 Exterior differentials

3.2 Constrained Optimization

4 Tangents and Multiplicities

4.1

5 Antiderivatives

6 Transcendental Functions